Entergy®

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Steve Vercelli
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10 CFR 50.73

RBG-48094

May 20, 2021

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2021-01-00, "Manual Reactor Scram due to Loss of Main Condenser Vacuum"

River Bend Station – Unit 1
NRC Docket No. 50-458
Renewed Facility Operating License No. NPF-47

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report. This document contains no commitments. If you have any questions, please contact Mr. Tim Schenk at 225-381-4177.

Respectfully,

SPV/twf

Enclosure: Licensee Event Report 50-458 / 2021-01-00, "Manual Reactor Scram due to Loss of Main Condenser Vacuum"

cc: NRC Regional Administrator - Region IV
NRC Project Manager - River Bend Station
NRC Senior Resident Inspector - River Bend Station
Louisiana Department of Environmental Quality
Public Utility Commission of Texas

Enclosure

RBG-48094

Licensee Event Report 50-458 / 2021-01-00, "Manual Reactor Scram due to Loss of Main Condenser Vacuum"

NRC FORM 366 (08-2020) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 08/31/2023
LICENSEE EVENT REPORT (LER) (See Page 3 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name	2. Docket Number	3. Page
River Bend Station – Unit 1	05000 458	1 OF 2

4. Title
Manual Reactor Scram due to Loss of Main Condenser Vacuum

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name: NA	Docket Number: 05000 NA
03	25	2021	2021	- 001 -	00	05	21	2021	Facility Name: NA	Docket Number: 05000 NA

9. Operating Mode	10. Power Level
1	80

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **Other** (Specify here, in Abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include Area Code)
Tim Schenk, Manager – Regulatory Assurance	225-381-4177

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
E	SH	CNV	N174	Y	NA	NA	NA	NA	NA

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ **Yes** (If yes, complete 15. Expected Submission Date)		NA	NA	NA

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On March 25, 2021 at 09:01 CDT, River Bend Station was operating at 93% reactor power when Main Condenser vacuum began to lower. Reactor power was lowered in an attempt to maintain Main Condenser vacuum. At 09:18 a manual reactor scram was initiated from approximately 80% reactor power due to Main Condenser vacuum continuing to lower. All control rods fully inserted and there were no complications. All systems responded as designed.

The loss of vacuum was caused by the in-service Steam Air Ejector Suction Valve closing due to faults within the valve's control components. The Steam Air Ejector Suction Valve function was restored by replacing the faulted components.

River Bend Station will establish a program to test the fail-as-is function of the Steam Air Ejector Suction Valves every outage.

LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
River Bend Station – Unit 1	05000- 458	2021	- 001	- 00

NARRATIVE

EVENT DESCRIPTION

On March 25, 2021 at 09:01 CDT, River Bend Station was operating at 93% reactor power when the Main Control Room Staff noted multiple trends of lowering Main Condenser (SG) vacuum, lowering generator (GEN) load, and lowering Offgas System (WF) flow. Reactor power was lowered in accordance with operating procedures in an attempt to maintain vacuum. The in-service Steam Air Ejector (SH) Suction Valve was found to have changed position from full open to full closed. A manual reactor scram was initiated at 09:18 due to Main Condenser vacuum continuing to lower. The scram was initiated at 80% reactor power. All control rods fully inserted, and all systems responded as designed.

Main Condenser vacuum was restored following the scram with the Condenser Air Removal Pumps (SH). Main Condenser vacuum did not lower to the automatic main steam line isolation (BL) setpoint. The Main Condenser was available throughout the event.

This event was reported under 10 CFR 50.72(b)(2)(iv)(B), as any event or condition that results in actuation of the Reactor Protection System when the reactor is critical and 10 CFR 50.72(b)(3)(iv)(A) Specified System Actuation as a result of expected post scram level 3 isolations. (EN 55154)

This report is made pursuant to 10 CFR 50.73(a)(2)(iv)(A), any event or condition that resulted in manual actuation of the Reactor Protection System.

SAFETY ASSESSMENT

This event did not result in any incremental risk or consequences to the general safety of the public, nuclear safety, industrial safety, or radiological safety. Main Condenser vacuum did not deteriorate to the Main Steam Isolation Valves automatic closure setpoint. Normal post-trip Main Condenser vacuum level was restored promptly by utilization of the Condenser Air Removal Pumps. Therefore, post-trip availability of the Main Condenser was not jeopardized.

EVENT CAUSE

The loss of Main Condenser vacuum was caused by the in-service Steam Air Ejector Suction Valve closing. Failure analysis following the event, revealed a fault in the valve controller and current-to-pressure (I/P) converter. The valve controller failure resulted in disruption of the electrical signal to the I/P converter. The I/P converter is designed to maintain the valve in position upon loss of electrical power or air pressure (fail-as-is). Because the I/P converter also failed, it did not lock the valve in the open position following failure of the controller.

The Steam Air Ejector Suction Valve design was modified to fail-as-is in September 2007. The recommended testing frequency for the new fail-as-is function was once per cycle. However, the recommended testing frequency was not implemented.

CORRECTIVE ACTIONS:

River Bend Station will establish a program to test the fail-as-is function of the Steam Air Ejector Suction Valves every outage.

PREVIOUS SIMILAR EVENTS

None.